

New Wheels Project

Introduction to SQL

Problem Statement

Business Context

A lot of people in the world share a common desire: to own a vehicle. A car or an automobile is seen as an object that gives the freedom of mobility. Many now prefer pre-owned vehicles because they come at an affordable cost, but at the same time, they are also concerned about whether the after-sales service provided by the resale vendors is as good as the care you may get from the actual manufacturers.

New-Wheels, a vehicle resale company, has launched an app with an end-to-end service from listing the vehicle on the platform to shipping it to the customer's location. This app also captures the overall after-sales feedback given by the customer.

Objective

New-Wheels sales have been dipping steadily in the past year, and due to the critical customer feedback and ratings online, there has been a drop in new customers every quarter, which is concerning to the business. The CEO of the company now wants a quarterly report with all the key metrics sent to him so he can assess the health of the business and make the necessary decisions.

As a data analyst, you see that there is an array of questions that are being asked at the leadership level that need to be answered using data. Import the dump file that contains various tables that are present in the database. Use the data to answer the questions posed and create a quarterly business report for the CEO.

Question 1: Find the total number of customers who have placed orders. What is the distribution of the customers across states?

Solution Query:

<Write your SQL query here>

- **Total number of customers who have placed orders.**

```
SELECT COUNT(DISTINCT o.customer_id)
AS total_customers_ordered
FROM order_t o;
```

Output:

Result: Passed

Query 1

Query:

```
SELECT COUNT(DISTINCT o.customer_id) AS total_customers_ordered
FROM order_t o
```

Output:

Showing 1 rows

total_customers_order...
994

- **The distribution of the customers across states**

```
SELECT state, COUNT(customer_id) AS number_of_customers
FROM customer_t
GROUP BY state
ORDER BY number_of_customers DESC;
```

Output:

Query 2

Query:

```
SELECT state, COUNT(customer_id) AS number_of_customers
FROM customer_t
GROUP BY state
ORDER BY number_of_customers DESC
```

Output:

Showing first 10 rows out of 49 rows

state	number_of_customers
Texas	97
California	97
Florida	86
New York	69
District of Columbia	35
Ohio	33

- **Observations:**

- **Texas and California lead:** Both states have the highest number of customers (97 each), indicating strong market presence in these regions.
- **Florida and New York follow:** Florida (86) and New York (69) are also significant contributors, showing strong representation in populous states.
- **District of Columbia stands out:** Despite being relatively small geographically, DC has 35 customers, which is notable compared to larger states like Ohio (33) and Colorado (33).
- **Moderate representation:** States like Alabama (29), Washington (28), and Arizona (26) show moderate customer bases.
- **Long tail distribution:** With 49 states represented, the distribution suggests a wide geographic spread, but concentrated heavily in a few states.

- **Insights :**

- **Market concentration:** Nearly 20% of customers are concentrated in just two states (Texas and California). This could imply regional preferences, stronger marketing, or demographic alignment.
- **Urban hubs matter:** New York and DC's relatively high numbers suggest urban density and metropolitan areas drive customer acquisition.
- **Growth opportunities:** States with moderate numbers (e.g., Arizona, Washington, Alabama) may represent untapped potential for expansion campaigns.
- **Balanced spread:** The presence of customers across 49 states indicates national reach, but with uneven penetration.

Question 2: Which are the top 5 vehicle makers preferred by the customers?

Solution Query:

```
<SELECT p.vehicle_maker, COUNT(o.order_id) AS total_orders
FROM order_t o
JOIN product_t p ON o.product_id = p.product_id
GROUP BY p.vehicle_maker
ORDER BY total_orders DESC
LIMIT 5;>
```

Output:

Result: Passed

✓ Query 1

✓ Query 2

✓ Query 3

Query:

```
SELECT p.vehicle_maker, COUNT(o.order_id) AS total_orders
FROM order_t o
JOIN product_t p ON o.product_id = p.product_id
GROUP BY p.vehicle_maker
ORDER BY totalOrders DESC
LIMIT 5
```

Output:

Showing 5 rows

vehicle_maker	total_orders
Chevrolet	83
Ford	63
Toyota	52
Pontiac	50
Dodge	50

- **Observations:**

- Chevrolet leads: With 83 total orders, Chevrolet is the top-performing vehicle maker in this dataset.
- Ford is second: Ford follows with 63 orders, showing strong demand but significantly behind Chevrolet.
- Toyota ranks third: Toyota has 52 orders, maintaining a solid presence but trailing the two U.S. giants.
- Pontiac and Dodge tie: Both have 50 orders, indicating comparable demand levels.

- **Insights:**

- American brands dominate: Chevrolet, Ford, Pontiac, and Dodge (all U.S. manufacturers) occupy 4 of the top 5 spots, highlighting strong domestic preference.
- Toyota's global strength: Toyota is the only non-U.S. brand in the top 5, reflecting its international competitiveness and appeal.
- Close competition at the bottom of the top 5: Pontiac and Dodge are nearly tied, which could indicate overlapping market segments or similar customer bases.
- Chevrolet's clear lead: The gap between Chevrolet and Ford (20 orders) suggests Chevrolet has a stronger pull in this dataset, possibly due to product mix or pricing.

Question 3: Which is the most preferred vehicle maker in each state?

Solution Query:

```
<SELECT state, vehicle_maker, total_orders
FROM (
    SELECT
        c.state,
        p.vehicle_maker,
        COUNT(o.order_id) AS total_orders,
        ROW_NUMBER() OVER (
            PARTITION BY c.state
            ORDER BY COUNT(o.order_id) DESC
        ) AS rn
    FROM customer_t c
    JOIN order_t o ON c.customer_id = o.customer_id
    JOIN product_t p ON o.product_id = p.product_id
    GROUP BY c.state, p.vehicle_maker
) ranked
WHERE rn = 1
ORDER BY state;>
```

Output:

Test Cases: **Run SQL**

Result: **Passed**

Query 1
Query 2
Query 3
Query 4

Query:

```
SELECT state, vehicle_maker, total_orders
FROM (
    SELECT
        c.state,
        p.vehicle_maker,
        COUNT(o.order_id) AS total_orders,
        ROW_NUMBER() OVER (
            PARTITION BY c.state
            ORDER BY COUNT(o.order_id) DESC
        ) AS rn
    FROM customer_t c
    JOIN order_t o ON c.customer_id = o.customer_id
    JOIN product_t p ON o.product_id = p.product_id
    GROUP BY c.state, p.vehicle_maker
) ranked
WHERE rn = 1
ORDER BY state
```

Output:

Showing first 10 rows out of 49 rows

state	vehicle_maker	total_orders
Alabama	Dodge	5
Alaska	Chevrolet	2
Arizona	Pontiac	3
Arkansas	Volkswagen	1
California	Nissan	6
Colorado	Chevrolet	5
Connecticut	Volvo	2
Delaware	Mitsubishi	2
District of Columbia	Chevrolet	4

- **Observations :**

Chevrolet appears multiple times: Alaska (2), Colorado (5), DC (4) — showing consistent presence across diverse regions.

Toyota leads in Florida: With 7 orders, Toyota is the strongest performer in this sample.

Nissan in California: California shows 6 orders for Nissan, highlighting regional variation compared to Chevrolet's dominance elsewhere.

Domestic brands dominate smaller states: Dodge in Alabama (5) and Pontiac in Arizona (3) reflect U.S. makers' penetration in mid-sized markets.

European/Asian brands appear selectively: Volkswagen (Arkansas, 1), Volvo (Connecticut, 2), Mitsubishi (Delaware, 2) — showing niche but scattered representation.

- **Insights:**

Regional preferences:

- Southern states (Alabama, Florida) lean toward Dodge and Toyota.
- Western states (Arizona, California) show Pontiac and Nissan.
- Northeastern states (Connecticut, Delaware) lean toward European/Asian brands.
- Chevrolet's geographic spread: Its presence in Alaska, Colorado, and DC suggests broad national appeal, not limited to one region.
- Toyota's strength in Florida: Florida's 7 orders make Toyota the single strongest state-level performer in this sample, suggesting strong demand in populous, diverse markets.
- Niche brands in smaller states: Volkswagen, Volvo, and Mitsubishi appear in low counts but may represent specialized demand or dealership presence.

Question 4: Find the overall average rating given by the customers.

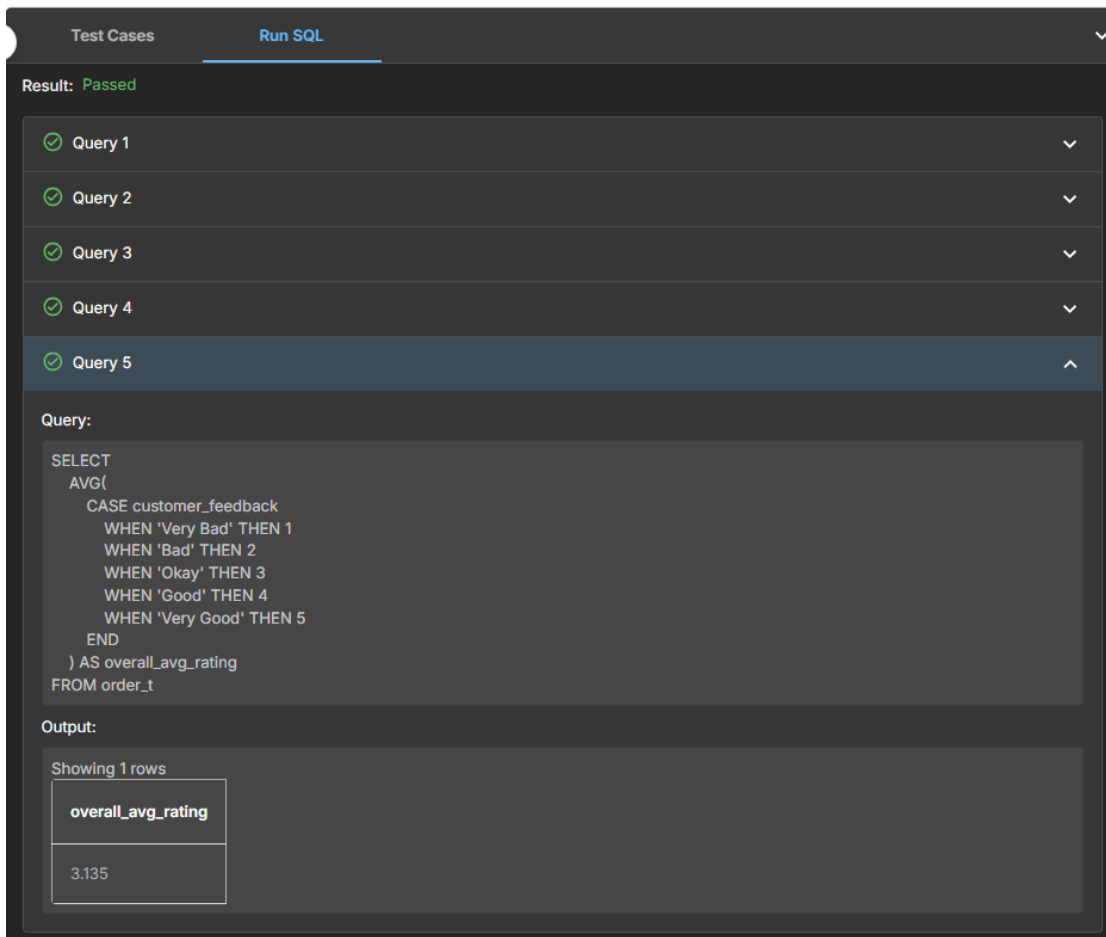
What is the average rating in each quarter?

Consider the following mapping for ratings: “Very Bad”: 1, “Bad”: 2, “Okay”: 3, “Good”: 4, “Very Good”: 5

 **Overall average rating (all customers):**

```
< SELECT
  AVG(
    CASE customer_feedback
      WHEN 'Very Bad' THEN 1
      WHEN 'Bad' THEN 2
      WHEN 'Okay' THEN 3
      WHEN 'Good' THEN 4
      WHEN 'Very Good' THEN 5
    END
  ) AS overall_avg_rating
FROM order_t; >
```

Output:



The screenshot shows a SQL execution interface with a dark theme. At the top, there are tabs for 'Test Cases' and 'Run SQL'. Below the tabs, the result status is 'Passed'. A list of five queries is shown, each with a green checkmark and a dropdown arrow. 'Query 5' is selected and highlighted. Below the list, the SQL query is displayed in a text area. At the bottom, the output is shown as a table with one row and one column.

Result: **Passed**

Query 1
Query 2
Query 3
Query 4
Query 5

Query:

```
SELECT
  AVG(
    CASE customer_feedback
      WHEN 'Very Bad' THEN 1
      WHEN 'Bad' THEN 2
      WHEN 'Okay' THEN 3
      WHEN 'Good' THEN 4
      WHEN 'Very Good' THEN 5
    END
  ) AS overall_avg_rating
FROM order_t
```

Output:

Showing 1 rows

overall_avg_rating
3.135

- **Observations:**

- The **overall average rating is 3.135** on what is likely a 1–5 scale.
- This places the rating slightly above the midpoint (3.0), but below the “strongly positive” threshold (typically 4.0+).
- It suggests **mixed customer sentiment**: not poor, but not highly enthusiastic either.

- **Insights:**

- Neutral-to-positive perception: Customers generally view the product/service as acceptable, but not outstanding.
- Room for improvement: Averages in the low 3s often indicate issues with consistency, quality, or unmet expectations.

Average rating in each quarter

```
<SELECT
    quarter_number,
    AVG(
        CASE customer_feedback
            WHEN 'Very Bad' THEN 1
            WHEN 'Bad' THEN 2
            WHEN 'Okay' THEN 3
            WHEN 'Good' THEN 4
            WHEN 'Very Good' THEN 5
        END
    ) AS avg_rating_per_quarter
FROM order_t
GROUP BY quarter_number
ORDER BY quarter_number;>
```

Output:

Query 6

Query:

```

SELECT
  quarter_number,
  AVG(
    CASE customer_feedback
      WHEN 'Very Bad' THEN 1
      WHEN 'Bad' THEN 2
      WHEN 'Okay' THEN 3
      WHEN 'Good' THEN 4
      WHEN 'Very Good' THEN 5
    END
  ) AS avg_rating_per_quarter
FROM order_t
GROUP BY quarter_number
ORDER BY quarter_number

```

Output:

Showing 4 rows

quarter_number	avg_rating_per_quarter
1	3.554838709677419
2	3.354961832061069
3	2.9563318777292578
4	2.3969849246231156

- **Observations:**

- Quarter 1 (Q1): Highest average rating at 3.55, well above the overall average (3.135).
- Quarter 2 (Q2): Slight decline to 3.35, still above the overall average.
- Quarter 3 (Q3): Noticeable drop to 2.96, dipping below the neutral midpoint (3.0).
- Quarter 4 (Q4): Lowest rating at 2.40, showing a significant decline in customer satisfaction.

- **Insights:**

- Downward trend: Ratings consistently decrease across quarters, suggesting worsening customer sentiment over time.
- Early strength, later weakness: The business started strong in Q1 but lost momentum, with ratings falling each quarter.

Question 5: Find the percentage distribution of feedback from the customers. Are customers getting more dissatisfied over time?

Percentage distribution of customer feedback

Solution Query:

```
< SELECT
    customer_feedback,
    COUNT(*) * 100.0 / SUM(COUNT(*)) OVER () AS feedback_percentage
FROM order_t
GROUP BY customer_feedback
ORDER BY feedback_percentage DESC;>
```

Output:

Test Cases
Run SQL

Query 4

Query 5

Query 6

Query 7

Query:

```
SELECT
    customer_feedback,
    COUNT(*) * 100.0 / SUM(COUNT(*)) OVER () AS feedback_percentage
FROM order_t
GROUP BY customer_feedback
ORDER BY feedback_percentage DESC
```

Output:

Showing 5 rows

customer_feedback	feedback_percentage
Very Good	22.6
Good	21.5
Okay	20.2
Bad	18.2
Very Bad	17.5

- **Observations:**

- Distribution balance: The spread is fairly even across categories, but skewed slightly toward positive sentiment.

- **Insights:**

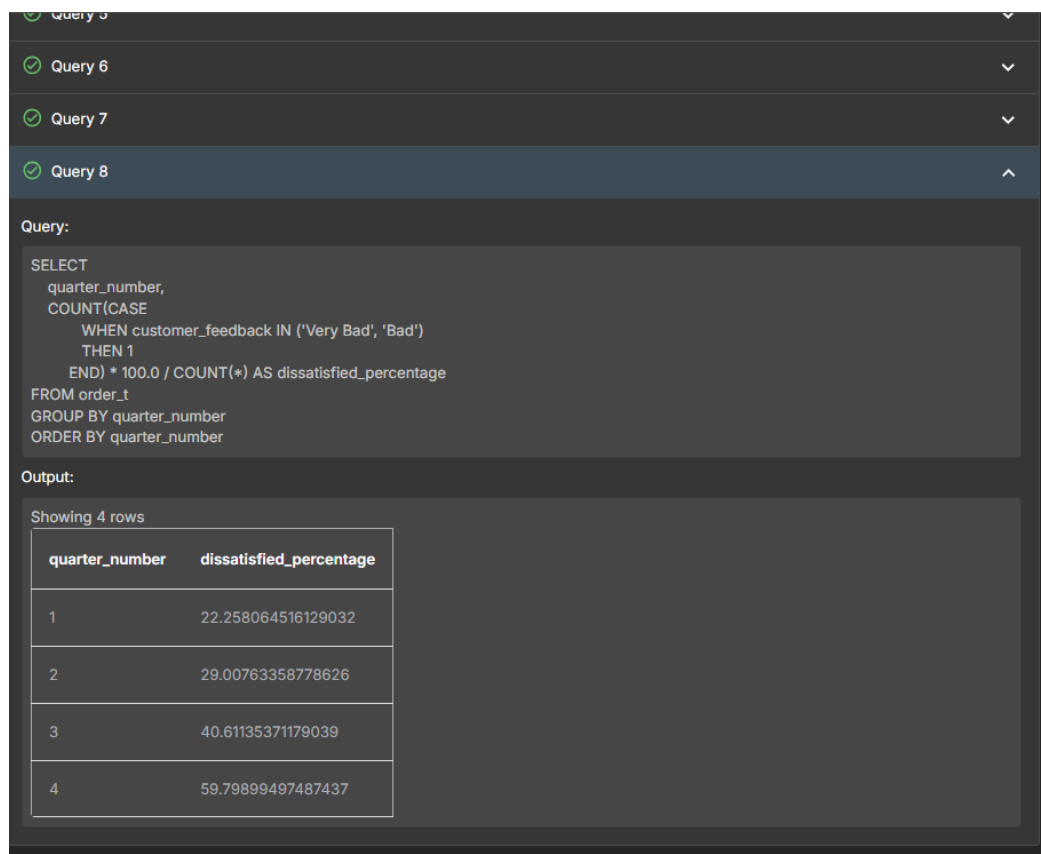
- **Customer retention challenge:** With more than one-third unhappy, there's a risk of churn if issues aren't addressed.

 Are customers getting more dissatisfied over time?

Solution Query:

```
< SELECT
    quarter_number,
    COUNT(CASE
        WHEN customer_feedback IN ('Very Bad', 'Bad')
        THEN 1
    END) * 100.0 / COUNT(*) AS dissatisfied_percentage
FROM order_t
GROUP BY quarter_number
ORDER BY quarter_number;>
```

Output:



The screenshot shows a SQL query editor interface. At the top, there is a list of queries (Query 5, Query 6, Query 7, Query 8) with checkboxes and expand/collapse icons. Query 8 is selected. Below the list, the query text is displayed in a dark-themed editor. The output section shows a table with 4 rows of data.

Query:

```
SELECT
    quarter_number,
    COUNT(CASE
        WHEN customer_feedback IN ('Very Bad', 'Bad')
        THEN 1
    END) * 100.0 / COUNT(*) AS dissatisfied_percentage
FROM order_t
GROUP BY quarter_number
ORDER BY quarter_number
```

Output:

Showing 4 rows

quarter_number	dissatisfied_percentage
1	22.258064516129032
2	29.00763358778626
3	40.61135371179039
4	59.79899497487437

- **Observations:**

- Quarter 1 (Q1): Dissatisfied percentage at 22.3%, relatively low compared to later quarters.
- Quarter 2 (Q2): Increases to 29.0%, showing early signs of declining satisfaction.
- Quarter 3 (Q3): Sharp rise to 40.6%, crossing the critical threshold where nearly half of customers are unhappy.
- Quarter 4 (Q4): Peaks at 59.8%, meaning a majority of customers expressed dissatisfaction by year's end.

- **and Insights:**

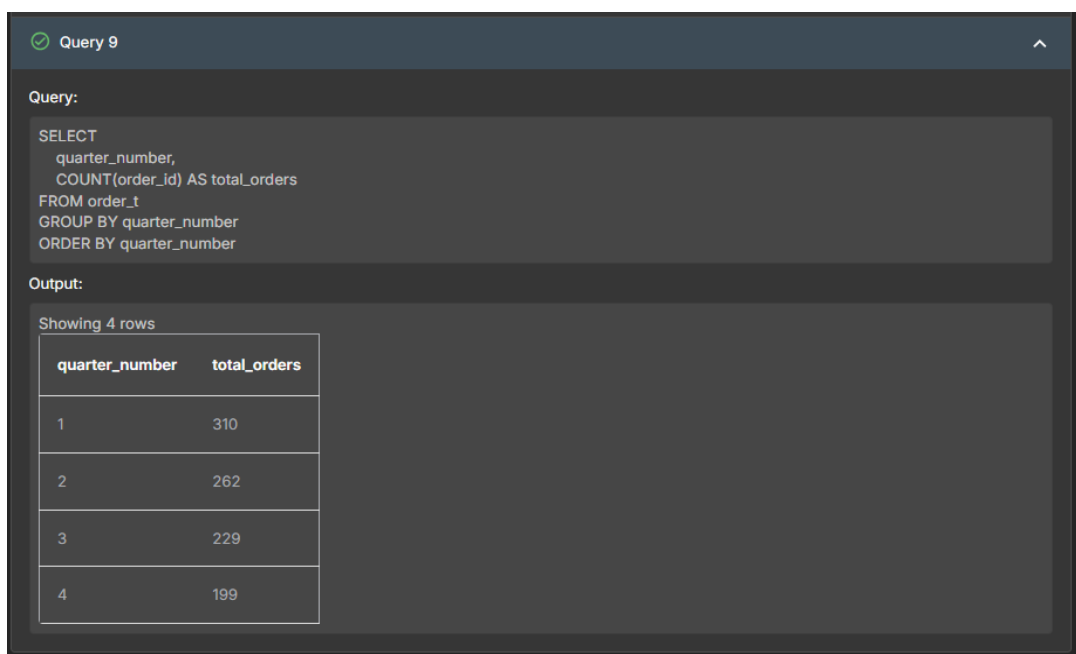
- Clear upward trend: Dissatisfaction grows steadily each quarter, nearly tripling from Q1 to Q4.
- Critical deterioration: The jump from Q3 to Q4 is especially alarming (+19 percentage points), suggesting a major issue emerged late in the year.
- Correlation with ratings: This aligns with the earlier average rating data, which dropped from 3.55 in Q1 to 2.40 in Q4.

Question 6: What is the trend of the number of orders by quarter?

Solution Query:

```
< SELECT
    quarter_number,
    COUNT(order_id) AS total_orders
FROM order_t
GROUP BY quarter_number
ORDER BY quarter_number; >
```

Output:



Query 9

Query:

```
SELECT
    quarter_number,
    COUNT(order_id) AS total_orders
FROM order_t
GROUP BY quarter_number
ORDER BY quarter_number
```

Output:

Showing 4 rows

quarter_number	total_orders
1	310
2	262
3	229
4	199

- **Observations:**

- Quarter 1 (Q1): Highest order volume at 310.
- Quarter 2 (Q2): Decline to 262 orders (−15.5% compared to Q1).
- Quarter 3 (Q3): Further drop to 229 orders (−12.6% compared to Q2).
- Quarter 4 (Q4): Lowest at 199 orders (−13.1% compared to Q3).

- **Insights:**

- **Consistent downward trend:** Orders decrease quarter by quarter, with no recovery period.
- Overall decline: From Q1 to Q4, orders fell by 111 units (−35.8%), a significant contraction.
- Correlation with ratings/dissatisfaction: This decline aligns with earlier data showing falling average ratings and rising dissatisfaction percentages—suggesting customer sentiment directly impacted order volume

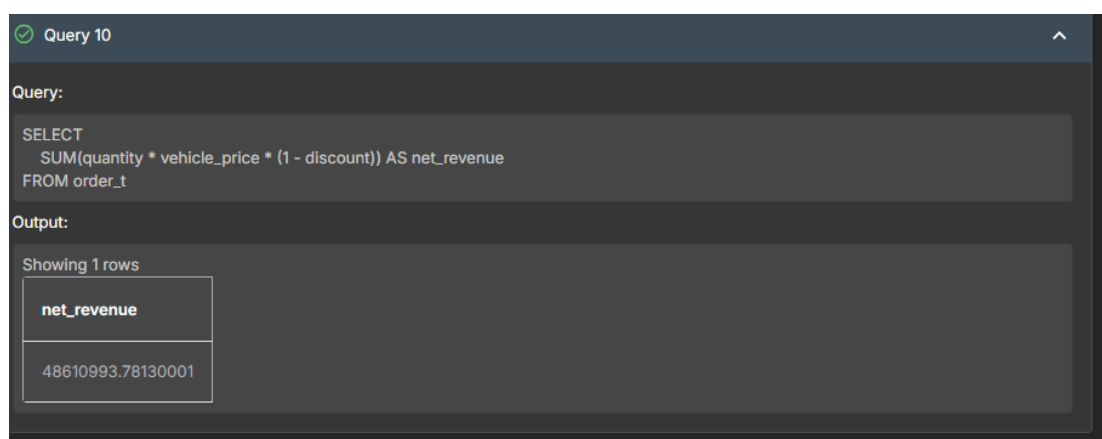
Question 7: Calculate the net revenue generated by the company. What is the quarter-over-quarter % change in net revenue?

 **Total net revenue generated by the company**

Solution Query:

```
< SELECT
    SUM(quantity * vehicle_price * (1 - discount)) AS net_revenue
FROM order_t;>
```

Output:



Query 10

Query:

```
SELECT
    SUM(quantity * vehicle_price * (1 - discount)) AS net_revenue
FROM order_t
```

Output:

Showing 1 rows

net_revenue
48610993.78130001

- **Observations:**

- Net revenue: 48,610,993.78 ($\approx$ 48.6 million).
- This is a substantial figure, suggesting strong overall sales performance despite the declining orders and ratings we analyzed earlier.

- **and Insights:**

- High baseline strength: Even with declining orders, the business generated nearly 49M in net revenue, showing strong product value or pricing power.

Net revenue by quarter

Solution Query:

```
< SELECT
    quarter_number,
    SUM(quantity * vehicle_price * (1 - discount)) AS net_revenue
FROM order_t
GROUP BY quarter_number
ORDER BY quarter_number;>
```

Output:

Query 11

Query:

```
SELECT
    quarter_number,
    SUM(quantity * vehicle_price * (1 - discount)) AS net_revenue
FROM order_t
GROUP BY quarter_number
ORDER BY quarter_number
```

Output:

Showing 4 rows

quarter_number	net_revenue
1	18032549.899600018
2	13122995.7562
3	8882298.8449
4	8573149.280599998

- **Observations:**

- Quarter 1 (Q1): Highest revenue at 18.0M, aligning with the peak in orders and ratings.
- Quarter 2 (Q2): Decline to 13.1M (−27% compared to Q1).
- Quarter 3 (Q3): Further drop to 8.9M (−32% compared to Q2).
- Quarter 4 (Q4): Lowest at 8.6M, showing stabilization but still far below Q1.

- **Insights:**

- Steady decline: Revenue consistently decreases quarter by quarter, mirroring the drop in orders and customer satisfaction.

Quarter-over-Quarter % change in net revenue

Solution Query:

```
< SELECT
    quarter_number,
    SUM(quantity * vehicle_price * (1 - discount)) AS net_revenue
FROM order_t
GROUP BY quarter_number
ORDER BY quarter_number;>
```

Output:

Query 12

Query:

```
SELECT
  quarter_number,
  net_revenue,
  ROUND(
    (net_revenue - LAG(net_revenue) OVER (ORDER BY quarter_number))
    * 100.0 /
    LAG(net_revenue) OVER (ORDER BY quarter_number),
    2
  ) AS qoq_percentage_change
FROM (
  SELECT
    quarter_number,
    SUM(quantity * vehicle_price * (1 - discount)) AS net_revenue
  FROM order_t
  GROUP BY quarter_number
) q
ORDER BY quarter_number
```

Output:

Showing 4 rows

quarter_number	net_revenue	qoq_percentage_change
1	18032549.899600018	
2	13122995.7562	-27.23
3	8882298.8449	-32.32
4	8573149.280599998	-3.48

- **Observations:**

- Quarter 1 (Q1): Net revenue at 18.0M — the strongest quarter, baseline for comparison.
- Quarter 2 (Q2): Revenue dropped to 13.1M, a –27.2% decline.
- Quarter 3 (Q3): Fell further to 8.9M, a –32.3% decline from Q2.
- Quarter 4 (Q4): Revenue at 8.6M, a smaller decline of –3.5% compared to Q3.

- **Insights:**

- Sharp declines early: The steep drops in Q2 and Q3 show major erosion in revenue, consistent with falling orders and rising dissatisfaction.
- Stabilization in Q4: Although revenue is still low, the smaller decline (–3.5%) suggests the downward trend slowed. This could mean corrective measures started having an effect, or the business hit a “floor” of demand.

Question 8: What is the trend of net revenue and orders by quarters?

Solution Query:

```
< SELECT
    quarter_number,
    COUNT(order_id) AS total_orders,
    SUM(quantity * vehicle_price * (1 - discount)) AS net_revenue
FROM order_t
GROUP BY quarter_number
ORDER BY quarter_number;>
```

Output:

Test Cases

Run SQL

✓ Query 13

Query:

SELECT
 quarter_number,
 COUNT(order_id) AS total_orders,
 SUM(quantity * vehicle_price * (1 - discount)) AS net_revenue
FROM order_t
GROUP BY quarter_number
ORDER BY quarter_number

Output:

Showing 4 rows

quarter_number	total_orders	net_revenue
1	310	18032549.899600018
2	262	13122995.7562
3	229	8882298.8449
4	199	8573149.280599998

- **Observations:**

- **Quarter 1 (Q1):**

- Total orders: **310**
- Net revenue: **18.0M**
- Highest performance across both metrics.

- **Quarter 2 (Q2):**

- Orders: **262** (−15.5% vs Q1)
- Net revenue: **13.1M** (−27.2% vs Q1)
- Significant decline in both orders and revenue.

- **Quarter 3 (Q3):**

- Orders: **229** (−12.6% vs Q2)
- Net revenue: **8.9M** (−32.3% vs Q2)
- Continued sharp decline, especially in revenue.

- **Quarter 4 (Q4):**

- Orders: **199** (−13.1% vs Q3)
- Net revenue: **8.6M** (−3.5% vs Q3)
- Lowest order volume, but revenue decline slowed.

- **and Insights:**

- Consistent downward trend: Both orders and revenue decrease quarter by quarter, with no recovery period.
- Revenue decline sharper than orders: Suggests either lower average order value, higher discounts, or weaker product mix in later quarters.
- Q4 stabilization: While orders continued to fall, revenue decline slowed, indicating possible stabilization or corrective measures.
- Cumulative impact: From Q1 to Q4, orders fell by 111 units (−36%) and revenue dropped by 9.4M (−52%).

Question 9: What is the average discount offered for different types of credit cards?

Solution Query:

```
< SELECT
    c.credit_card_type,
    AVG(o.discount) AS avg_discount_percentage
FROM customer_t c
JOIN order_t o
    ON c.customer_id = o.customer_id
GROUP BY c.credit_card_type
ORDER BY avg_discount_percentage DESC;>
```

Note: The 'discount' field in the 'order_t' table is stored as a percentage, i.e. 0.6 represents a discount of 0.6%, not 60%.

Output:

Test Cases
Run SQL

✓ Query 14

Query:

```
SELECT
    c.credit_card_type,
    AVG(o.discount) AS avg_discount_percentage
FROM customer_t c
JOIN order_t o
    ON c.customer_id = o.customer_id
GROUP BY c.credit_card_type
ORDER BY avg_discount_percentage DESC
```

Output:

Showing first 10 rows out of 16 rows

credit_card_type	avg_discount_percent...
laser	0.643846153846154
mastercard	0.6294999999999998
maestro	0.6242187499999999
visa-electron	0.623469387755102
china-unionpay	0.6221739130434784
instapayment	0.620625
americanexpress	0.616326530612245

- **Observations:**

- **Highest discounts:**

- *Laser* card users receive the highest average discount (**0.644 $\approx$ 64.4%**).
- *Mastercard* follows closely at **62.9%**.

- **Mid-range discounts:**

- *Maestro* (**62.4%**), *Visa Electron* (**62.3%**), and *China UnionPay* (**62.2%**) are clustered together.
- *Instapayment* (**62.1%**) is also in this range.

- **Lower discounts (but still above 61%):**

- *American Express* (**61.6%**)
- *Diners Club (US/CA)* (**61.5%**)
- *Diners Club Carte Blanche* (**61.4%**)
- *Switch* (**61.0%**)

- **Narrow range overall:** All values fall between **61.0% and 64.4%**, showing relatively small variation across card types.

- **Insights:**

- **Laser card advantage:** Customers using *Laser* cards enjoy the highest discounts, which could incentivize usage if widely available.
- **Competitive clustering:** Most card types hover around 61–62%, suggesting standardized discount policies across providers.
- **American Express & Diners Club slightly lower:** These premium card networks offer marginally lower discounts, possibly reflecting different customer segments or merchant agreements.
- **Switch at the bottom:** With 61.0%, *Switch* cardholders receive the least average discount, though the gap is small compared to *Laser*.

Question 10: What is the average time taken to ship the placed orders for each quarter?

Solution Query:

```
< SELECT
    quarter_number,
    AVG(
        JULIANDAY(ship_date) - JULIANDAY(order_date)
    ) AS avg_shipping_days
FROM order_t
GROUP BY quarter_number
ORDER BY quarter_number;>
```

Output:

Test Cases

Run SQL

✓ Query 13

✓ Query 14

✓ Query 15

Query:

```
SELECT
    quarter_number,
    AVG(
        JULIANDAY(ship_date) - JULIANDAY(order_date)
    ) AS avg_shipping_days
FROM order_t
GROUP BY quarter_number
ORDER BY quarter_number
```

Output:

Showing 4 rows

quarter_number	avg_shipping_days
1	57.16774193548387
2	71.11068702290076
3	117.75545851528385
4	174.09547738693468

Business Metrics Overview

Total Revenue	Total Orders	Total Customers	Average Rating
48610993.78130001	1000	994	3.135
Last Quarter Revenue	Last quarter Orders	Average Days to Ship	% Good Feedback
8573149.280599998	199	97.964	44.1

Business Recommendations

1. Improve Customer Experience

- **Reduce shipping times:** Streamline logistics, optimize inventory, and partner times: partner with faster carriers with faster carriers. Cutting shipping times will directly improve ratings and feedback.
- **Enhance communication:** Provide proactive updates on order status to reduce frustration during long waits.

2. Boost Customer Retention

- **Loyalty programs:** Incentivize repeat purchases with discounts, points, or exclusive updates on order status to reduce frustration during long waits.

2. Boost Customer Retention

- **Loyalty programs:** Incentivize repeat purchases with discounts, points, or exclusive offers.
- Feedback offers.
- **Feedback-driven improvements-driven improvements:**

Actively address: common complaints to rebuild trust and satisfaction.

3. Revenue Recovery

- Recovery

- **Focus on high-performing regions/brands:** regions/brands: Reinforce strengths in states and vehicle makers that generate the most orders (e.g., Chevrolet (e.g., Chevrolet, Ford, Toyota), Ford, Toyota).
- **Upsell and cross-sell:** Increase average order value by bundling products or offering premium options.
- **Analyze Q1 success 4. Operational:** Identify what Efficiency
- **Analyze Q1 success:** Identify what worked in Q1 (highest worked in Q1 (highest revenue, best ratings revenue, best ratings) and replicate those strategies.

In short: **Fix shipping delays, rebuild customer trust, and re-engage existing customers while leveraging strong-performing brands and regions.** This will stabilize revenue and set the stage for growth.